

Hazard Red Zone

Dashboard pe jo dikhta hai, woh data kahan se aata hai — live, file, demo CSV, ya app ne calculate kiya. Friend / teammate share guide.

1. Labels (pehle yeh yaad rakho)

Label	Matlab
LIVE	Abhi internet se aa raha hai (Open-Meteo, Twilio, browser GPS, OSM tiles).
REAL FILE	Asli public dataset repo mein hai (Kaggle flood/landslide). Roz naya NDMA feed nahi.
DEMO CSV	Humne Kerala-style seed likha. Official live district dashboard nahi.
COMPUTED	App / Random Forest / tumhari click se bana (score, count, order id).
HARDCODED UI	Text/layout code mein likha. Database nahi.

UI data **GET /api/...** se aata hai. Backend **ai_model/*.csv**, Python models, aur Open-Meteo padhta hai.

2. Login

Jo dikhta hai	Kahan se	Type
Collector / Field / Resident cards	Login.jsx text	HARDCODED UI
Session name, office	POST /api/login — fixed desks, no password	HARDCODED

3. Overview — upar ke 5 numbers

UI	Source	Type
Habitations at risk	Habitation list length (6 sites)	COMPUTED
Critical / Immediate	Jahan riskLevel = CRITICAL	COMPUTED (model)
Still need relocation	Status PENDING / VERIFIED count	COMPUTED + verify/approve
Safe-site capacity	Camps ke capacity ka sum	DEMO CSV relief_camps_kerala.csv
Currently sheltered	Camps ke occupied ka sum	DEMO CSV
Active / available teams	Team status	DEMO CSV + dispatch click
Families in camps / reunited	Reunification records	DEMO CSV + jo tum register karo

Hint “AI-ranked from live telemetry” tab sach hai jab Overview pe **LIVE** on ho.

4. Overview — pipeline aur do tracks

Saara copy PPT wala story hai. **HARDCODED UI** — koi API table nahi.

5. Overview — Kerala map

Layer	Source	Type
Background (roads, nadi, towns)	OpenStreetMap tiles	LIVE (API key nahi)
Red risk dots + circles	6 habitations: names/coords in real_data_sampler.py; colour = model score	Real Kerala lat/lng + COMPUTED risk
Green camps	relief_camps_kerala.csv lat/lng	DEMO CSV (coords Kerala ke)
Purple incidents	active_incidents.csv	DEMO CSV (rain numbers Kaggle row se)
Orange SOS	Citizen SOS form	User action; pin = browser GPS ya Aluva 10.1077, 76.3546

Map model ko **nahi** jaata. Sirf dikhana hai.

6. Overview — weather

UI	Source	Type
LIVE button	Har habitation ke lat/lng pe Open-Meteo → flood/landslide features → Random Forest rescore	LIVE
Rain, temp, humidity, soil moisture	Open-Meteo forecast (free, no key)	LIVE
River discharge	Open-Meteo Flood API, jahan cell mile	LIVE (warna estimate)
Water level (m)	CWC gauge nahi; rain + discharge se formula	COMPUTED / estimated
NORMAL / SEVERE MONSOON / CYCLONE	Code: 35 / 310 / 440 mm etc.	HARDCODED demo

7. Overview — incidents + model evidence

UI	Source	Type
#SIH-090 Flash Flood, location...	active_incidents.csv	DEMO CSV
Team assigned	CSV seed + Dispatch click	DEMO + COMPUTED
SOS lines	Memory / backend/data/state.json	User LIVE action

Flood rows ~10000, positives	flood_risk_dataset_india.csv count	REAL FILE
Landslide rows ~5000, high-risk	regenerated_landslide_risk_dataset.csv	REAL FILE

8. Relocation plan

Column	Source	Type
Name (Munnar, Aluva...)	real_data_sampler.py list	Hardcoded names, real coords
Population	Same list (450, 1200...)	DEMO
Hazard type	Same list	DEMO
Flood / landslide / risk score	Random Forest .pkl	COMPUTED — inputs LIVE Open-Meteo ya scenario ya pehle Kaggle row
Priority Immediate / Short / Medium	Score rule (≥ 75 Immediate)	COMPUTED
Matched safe site, km, capacity	Sampler pairing	DEMO (live matcher nahi)
Capacity OK	cap $\geq$ population	COMPUTED
PENDING / VERIFIED / APPROVED	Memory + state.json	COMPUTED (clicks)
ORD-2026-xxxx	Approve button	COMPUTED

Verify GPS: pehle **live browser GPS try**, fail ho to habitation lat/lng. Photo: sirf **filename**, image upload nahi.

9. Field notes

UI	Source	Type
Task list	Backend seed — top habitations	COMPUTED (wahi 6 sites)
Checklist ticks	Officer click	Session LIVE
Officer name	Default "SDRF..."	HARDCODED default
Photo	File ka naam only	Dummy upload
GPS in API	Habitation centroid	Is screen pe live GPS nahi
Offline queue	browser localStorage	Device pe

10. Alerts

UI	Source	Type
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Habitation / shelter dropdown	Habitations + matched site	DEMO list
Message	Default English, editable	HARDCODED default
SMS	Twilio if backend/.env set	LIVE or simulator
IVR	Browser speechSynthesis	Voice LIVE, carrier nahi
App / siren	Log only	DEMO
Alert log	Memory + state.json	Jo tumne bheja

11. Rescue dispatch

UI	Source	Type
Incidents	active_incidents.csv + SOS	DEMO + user SOS
Rain/water in description	CSV (Kaggle row copy)	REAL FILE numbers, live gauge nahi
NDRF / SDRF teams	rescue_teams_india.csv	DEMO CSV
Dispatch Active	Button click	COMPUTED
A* path, safety, ETA	ai_model/astar.py 20×20 grid	COMPUTED — OSM roads nahi

12. Families + Resident desk

UI	Source	Type
Aarav, Ananya...	family_reunification.csv	DEMO people
Counts	Unhi records + naye	COMPUTED
Register / reunite	Form → memory / state.json	Session LIVE
Camps on citizen page	relief_camps_kerala.csv	DEMO CSV
SOS defaults	"Aluva Bridge Underpass", 2 people	HARDCODED
SOS map pin	Browser GPS or Aluva fallback	Mix
Official alerts	Alert log	Jo authority ne bheja

13. Model — train vs predict

Train (ek baar, historical)

- flood_risk_dataset_india.csv (~10,000 rows) → flood_prediction_rf_model.pkl
- regenerated_landslide_risk_dataset.csv (~5,000 rows) → landslide_prediction_rf_model.pkl

Predict (ab, har score)

- **Static:** lat/Ing, soil, land cover, population (sampler + Kaggle row)
- **Dynamic:** LIVE Open-Meteo, ya NORMAL/MONSOON/CYCLONE buttons

14. Judges / friend ke liye 4 line

1. Map background **live OSM**; pins hamari files + model.
2. Risk score **asli trained Random Forest**; mausam **LIVE Open-Meteo** (ya demo cyclone).
3. Camps, teams, families, incidents **demo CSV** — Kerala coords, live NDMA nahi.
4. SOS, approve, dispatch, alerts **is session ke real actions** (SMS Twilio optional).

Project: Hazard Red Zone Alert & Rescue Coordination System · Problem SIH26191

File generated for sharing — not an official NDMA dataset document.